PX-400 Motion Controller

Hardware Design Specification, Board PCB-400 Rev C

Oriole Packaging Systems — Hardware Engineering

Revision C August 2026 CONFIDENTIAL

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1 Overview

Board PCB-400 Rev C is the motion controller for the PX-400 packaging system. It drives one index axis, samples a differential encoder channel, and reports faults including E-217 (encoder signal loss during index). This specification covers the encoder interface, signal-integrity margins, and the interaction with firmware diagnostic timing.

2 Encoder interface

The encoder connects at J7 (17-pin circular). The interface uses RS-422 differential pairs with 120 ohm termination at the controller receiver. The cable shield drain terminates at the controller end only; terminating both ends creates a ground loop that injects index-synchronous noise. The receiver requires a minimum differential amplitude of 200 mV throughout the index cycle.

3 Signal integrity and firmware 4.8 timing

Firmware 4.8 tightened the E-217 diagnostic sampling window to catch genuine signal dropouts earlier. Internal testing showed the tighter window also exposes marginal field connections: oxidized J7 contacts or fatigued shield terminations that previously recovered within the old window now trip E-217. The failure signature is intermittent stops correlated with the firmware 4.8 deployment date, D-17 passing on a clean bench but failing on the line, and no failed component on the bench.

Parameter	Design margin	Field finding
Differential amplitude	$>$ 200 mV required, 480 mV typical	Marginal below 260 mV on aged J7
Dropout tolerance	2 ms (firmware 4.8)	Old window masked up to 6 ms
Shield termination	Controller end only	Double-terminated sites noisier

The approved field response is connector and harness inspection with D-17, not component replacement and not firmware downgrade. Firmware 4.8.1 relaxes the sampling window and extends D-17 sampling; see the release note.

4 Approved field summary

Field service may be told: E-217 after firmware 4.8 usually indicates a marginal encoder connection exposed by tighter diagnostics; inspect J7 and the harness, run D-17, capture logs, replace EA-400 or HN-44 only on diagnostic evidence. The contents of this specification beyond that summary stay confidential.

5 Revision history

Rev	Date	Change
B	2025-03	Initial release
C	2026-08	Added Section 3 timing analysis after field E-217 cluster